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Miocene paleomagnetic results from southeastern Korea

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Abstract

Paleomagnetic samples were collected from 25 sites of Late Oligocene to Middle Miocene age from southeastern South Korea. The mean direction obtained is $D = 39.4°$, $I = 46.2°$ with $\alpha_{95} = 9.5°$, which is statistically distinguishable from the present axial geocentric dipole field direction. When combined with the previous reported results of eight sites from this region, we get a mean direction ($D = 40.1°$, $I = 47.4°$, $\alpha_{95} = 7.9°$) for 33 sites of Late Oligocene to Middle Miocene age. This direction yields an early Neogene virtual geomagnetic pole of 56.5°N, 215.2°E with an α_{95} of 7.9°, discordant from the Cretaceous reference pole for Korea. The paleomagnetic data suggest that portions of southeastern Korea underwent a large clockwise rotation of about 40° during the Neogene. Two large NE-trending right-lateral faults (the Yangsan Fault and the Dongnae Fault, Fig. 1) cut the study areas. Both these faults show evidence for recent activity. In addition, minor faults with similar trends to these major faults occur. The observed deflections in paleomagnetic declinations were probably caused by rigid block rotations of crustal blocks in a right lateral shear system during the Neogene. The paleomagnetic directions for Miocene aged rocks from southeastern Korea and southwestern Japan are similar in direction and magnitude. This suggests that the cause of the rotations in both areas may be related to a similar tectonic event in this area during the Neogene.

1. Introduction

There have been several paleomagnetic studies on Korea with special reference to plate tectonic reconstructions of this area.

Yaskawa (1975) combined paleomagnetic results from Cretaceous aged rocks from South Korea (Kienzle and Sharon, 1966) and results from a Cretaceous sandstone from North Korea (Gurarii et al., 1966). From the data he obtained a Cretaceous pole for Korea of 72.0°N, 210.6°E. Ito and Tokieda (1980) reported results from Late Cretaceous granites. However, all of their samples have single polarities (all normal) and lack structual corrections, which makes it difficult to evaluate their reported pole. Kang and Lee (1972) studied Early Cretaceous intrusive rocks collected from various locations in North Korea. Their reported Cretaceous pole of 70°N, 186°E is indistinguishable from Lin's (1984) Early Cretaceous pole reported for northern China (69°N, 200°E, $\alpha_{95} = 12°$).

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Three groups of researchers have studied the Cretaceous Gyongsang Basin of southeastern Korea. Otofuji et al. (1983) showed a pole for South Korea which is similar to results from China, suggesting that Korea and China are part of the same tectonic block since the Cretaceous. Kim and Jeong (1986) studied Cretaceous rocks from the Gyeongsang Basin and they also concluded that the Cretaceous pole for Korea is indistinguishable from Cretaceous poles obtained for East Asia. Lee et al. (1987) found a stable primary magnetization of high blocking temperatures in Early Cretaceous rocks from this portion of Korea. Their study yields a Early Cretaceous pole of 68°N, 205°E with an α_{95} of 6° which is coincident with the Early Cretaceous pole for China (Lin et al., 1985).

Kienzle (1968) studied Tertiary volcanic rocks cropping out along the southeastern margin of Korea (Ulenung Island and Jeju Island). Because Kienzle (1968) used low demagnetization fields (20 mT or less) and showed no vector plot of the demagnetization, it is difficult to evaluate if he was able to isolate a primary magnetization from NRM. Won et al. (1986) and Kim and Jeong (1986) reported paleomagnetic results from olivine basalts (from Pliocene to Holocene) from Jeju Island. Both studies showed Plio-Pleistocene poles which are indistinguishable from the present geomagnetic dipole field direction. A similar paleomagnetic Plio-Pleistocene pole has also been reported from basalts from a volcanic region north of Seoul (Kim et al., 1984) and from basalts cropping out in North Korea (Kang, 1966).

Kim et al. (1986) conducted paleomagnetic and K-Ar studies on 42-Ma porphyries and Late Oligocene–Early Miocene volcanic rocks in southeastern Korea. In contrast to the paleomagnetic results from late Cenozoic rocks from other regions in Korea, Kim et al. (1986) reported paleomagnetic declinations which are deflected more than 40° clockwise from the present axial geocentric dipole field direction. Paleomagnetic directions obtained from Eocene porphyries are indistinguishable from Late Oligocene to Early Miocene volcanic rocks from southern Korea. The data from this region suggest that since the Early Miocene southeastern Korea has undergone a relatively local clockwise rotation. Kim and Van der Voo (1990) also showed the Triassic paleomagnetic directions from the Ryongam Massif and its border area, South Korea. Their mean direction ($D = 62.9°$, $I = 35.7°$, $\alpha_{95} = 15°$) is also deflected east-northeasterly and interpretated as a result of relative local rotations of the blocks about vertical axes. Contrary to the Early Neogene clockwise rotations by Kim et al. (1986) and Kim and Van der Voo (1990), paleomagnetic results from Eocene, Oligocene and Miocene basalts collected from North Korea showed declinations deflected over 20° counterclockwise from the dipole field direction.

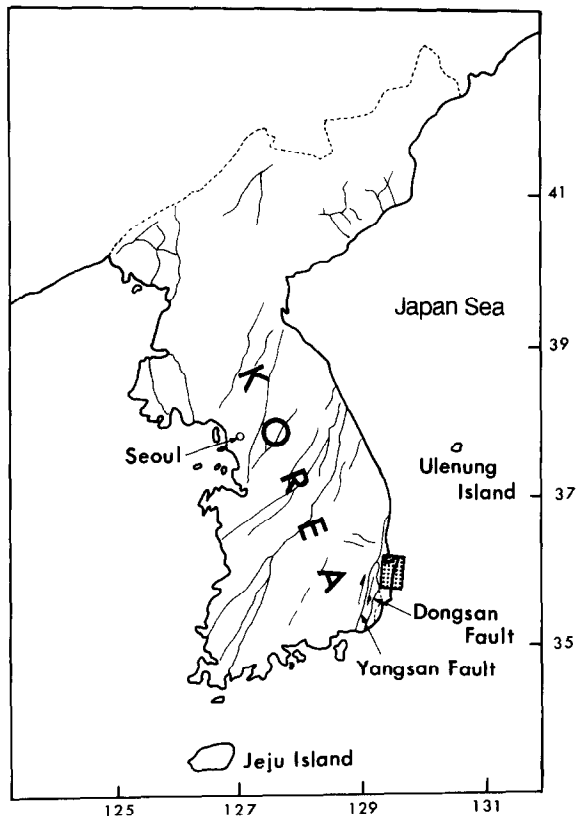


Fig. 1. Simplified tectonic map of Korea showing location of study area and large right-lateral strike-slip faults.

In this paper, we will present a new data set obtained from 25 volcanic flows in southeastern Korea (Fig. 1) and discuss them in the light of a local rotation in a large right-lateral shear zone.

2. Geology of the sampling region

Most of the Korean Peninsula is underlain by Precambrian to Mesozoic rocks of continental origin. In South Korea, these older rocks are exposed in numerous NNE-trending belts. Many of these belts are mapped as fault bounded (Korean Institute of Energy and Resources, 1981). In southeastern Korea, pre-Cretaceous rocks are unconformably overlain by two NE-trending basins, the Cretaceous Gyeongsang Basin and the middle to late Tertiary Pohang Basin (Chang, 1975). The Pohang Basin is located along the eastern margin of South Korea. This basin is composed of marine sediments and mafic and intermediate volcanic flows and tuffs. Biostratigraphic data from the sediments suggest that the basin began to form in the Late Oligocene to Miocene (Chang, 1975).

Large NNE-trending strike-slip faults cut Neogene strata along the eastern margin of Korea and in the Uleung (Tsushima) Basin, east off the Korean Peninsula (Lee et al., 1984). Geological-geophysical data from this area indicate that right-lateral motion with about 25 km displacement has occurred on these NE–trending faults (Um et al., 1983). While the timing of motion is uncertain, numerous historic earthquakes and microseismic activity along these faults suggest that they are currently active (Lee and Na, 1983; Kim, 1987). Map relations show faults cutting Neogene strata in the Pohang Basin (Chwae et al., 1988). The map relations and the active seismicity in the region indicate that the faulting system in this region has been active during the Neogene.

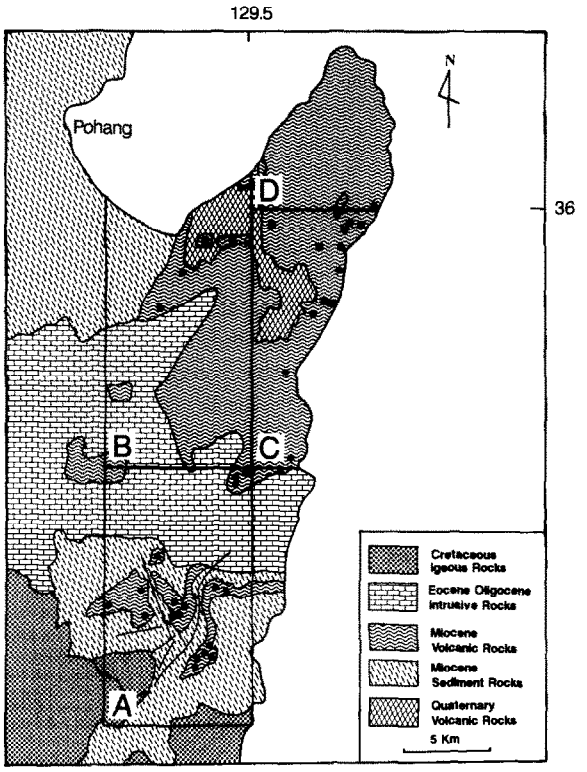


Fig. 2. Map of Pohang area showing the locations of sampling sites and specific area in this study. Sampling sites from this study and Kim et al. (1986) are shown with ● and ■, respectively. *A* = Eoil area; *B* = Yongduk area; *C* = Guryongpo area; *D* = Daebo area.

3. Experimental procedures

A total of 148 independently oriented samples were collected from 25 sites in middle Tertiary volcanics from the Pohang Basin (Fig. 2). Each site reported in Table 1 represents a different lava flow or tuffaceous unit and, therefore, corresponds to a distinct time unit. Ages assigned to the sites are based on previously reported K-Ar dating (Kim et al., 1986; Jin et al., 1988), micropaleontological analyses of sedimentary rocks interbedded with volcanic rocks (Yoon, 1975; Lee, 1975; Kim, 1986), and our own K-Ar dates. Except for one anomalous K-Ar age of 10 Ma, the other ten radiometric ages reported from this region and our two additional K-Ar isotopic ages (obtained commercially from Geochron) yield restricted ages between 22 and 15 Ma.

Tilting of rocks in this region is gentle. Many of the tuffaceous units observed in the field are horizontal. This statement is based on horizontal sedimentary rocks and volcanic tuffs, vertical columnar joints, and/or horizontal baked contacts, which we observed in close proximity to the collection sites. Tilt correction was made on seven sites on the basis of the tilt of the interbedded sedimentary units measured at or near the collection site. The sites which showed a tilt, were gently dipping (usually less than 20°). Although

there is a very small increase in the precision parameter κ after bedding correction (from 9.5 to 10.3), the fact that the rocks in this region are horizontal or only gently dipping made us unable to perform fold tests on the data.

Four to eight cylindrical samples (25 mm in diameter) were drilled at each site using a hand-held, gasoline-powered portable rock drill. These cores were oriented by an orienting device and a Brunton compass. At three sites, seven oriented hand samples were collected for each site. One specimen from each oriented hand sample was used in calculating mean directions.

The volcanic rocks studied contained a strong magnetization that was easily measured using a spinner magnetometer (Minispin by Molspin Ltd.). Stepwise alternating field (AF) demagnetization was carried out on two to four pilot specimens from each site using a Schonstedt GSD-1 single-axis demagnetizer. Orthogonal vector plots were constructed to observe the behavior of the remanent magnetization during progressive demagnetization (Zijderveld, 1967). A characteristic component of magnetization for each specimen was considered isolated when the orthogonal projections showed the magnetization components decaying in a unidirectional fashion towards the origin with progressive treatment and when less than 10% of the initial magnetization remained (Fig. 3a). Stepwise thermal demagnetization techniques were also employed on samples from each site. Stability against both AF and thermal demagnetization is generally good (Fig. 3a and b). Even when the NRM includes a significant sec-

Table 1
Miocene paleomagnetic results from southeastern Korea

Site	N	Locality		Rock type	Demag AF/TH	Dec	Inc	α_{95}	k	VGP	
		Lat (°N)	Long (°E)							Lat (°N)	Long (°E)
KK-1	6	35.80	129.39	tuff	40/–	182.7	−34.9	6.3	114.5	73.3	300.5
KK-3	4	35.81	129.41	tuff	10/–	45.5	45.5	14.2	42.6	50.2	220.3
KK-6	6	35.83	129.42	basalt	30/–	42.2	52.0	9.7	48.5	55.2	212.2
KK-9	5	35.80	129.44	basalt	30/–	231.9	−3.4	6.3	147.8	31.2	242.6
KK-18	4	35.78	129.45	basalt	30/–	245.4	−7.1	10.9	71.5	21.9	231.4
KK-20	7	35.78	129.46	tuff	/300	28.5	40.7	4.4	161.7	62.4	238.5
KK-26	6	35.87	129.50	basalt	20/–	57.1	60.7	4.2	257.5	45.9	193.8
KK-36[a]	5	36.01	129.58	andesite	30/–	17.2	27.4	6.0	161.1	63.6	269.5
KK-43	5	35.98	129.46	tuff	30/–	10.5	54.7	8.5	82.1	81.4	221.4
KK-44	6	35.97	129.55	andesite	30/–	28.7	42.6	11.7	34.0	62.9	236.1
KK-46	6	35.88	129.51	basalt	20/–	44.9	54.0	3.5	370.2	53.6	208.1
KK-47	5	35.87	129.49	basalt	30/–	69.9	56.0	2.5	945.6	34.9	196.5
KK-48	6	35.87	129.49	basalt	30/–	48.4	60.3	2.8	586.0	52.3	196.2
KK-49 [a]	5	35.80	129.43	tuff	10/–	206.5	−54.0	2.6	874.5	68.4	215.0
KK-55	8	35.80	129.41	tuff	–/300	228.1	−80.3	2.9	369.3	46.8	150.0
K-1	5	35.80	129.44	basalt	40/–	44.1	55.5	8.3	85.7	54.6	205.7
K-2	7	35.79	129.46	basalt	30/–	16.0	6.5	3.3	341.8	54.3	281.3
K-3 [a]	5	35.81	129.47	basalt	30/–	256.8	−23.1	5.8	173.1	17.6	217.2
K-5 [a]	7	35.87	129.49	basalt	40/–	49.3	53.9	5.4	123.9	50.1	206.6
K-7	7	35.92	129.51	basalt	30/–	223.4	−44.3	5.2	135.7	51.8	222.1
K-8	6	35.95	129.53	basalt	40/–	201.0	−7.6	4.4	228.4	52.5	273.6
K-11	6	35.97	129.45	basalt	–/300	197.3	−74.7	9.4	51.4	62.4	147.4
K-12	7	35.98	129.46	basalt	30/–	197.9	−69.2	4.9	154.8	69.0	160.7
K-13	6	36.01	129.48	basalt	30/–	247.2	−49.7	3.4	397.1	34.7	204.4
K-14 [a]	7	35.95	129.44	tuff	20/–	43.5	44.6	9.7	39.6	51.8	221.7

N = number of samples; Demag = demagnetization method, AF = alternating field demagnetization level used to obtain characteristic direction given in mT, TH = thermal demagnetization (temperature is given in degrees centigrade); D = mean declination; I = mean inclination; α_{95} = radius of 95% confidence circle; k = precision parameter.
[a]Bedding corrected.

ondary component, we succeeded in obtaining the stable magnetic vector (Fig. 3c). In all cases the mean direction obtained after thermal and AF demagnetizations were indistinguishable. The characteristic directions from each site were combined into a site mean direction. The data from individual sites are shown in Figs. 4 and 5 and listed in Table 1. Because the sites reported

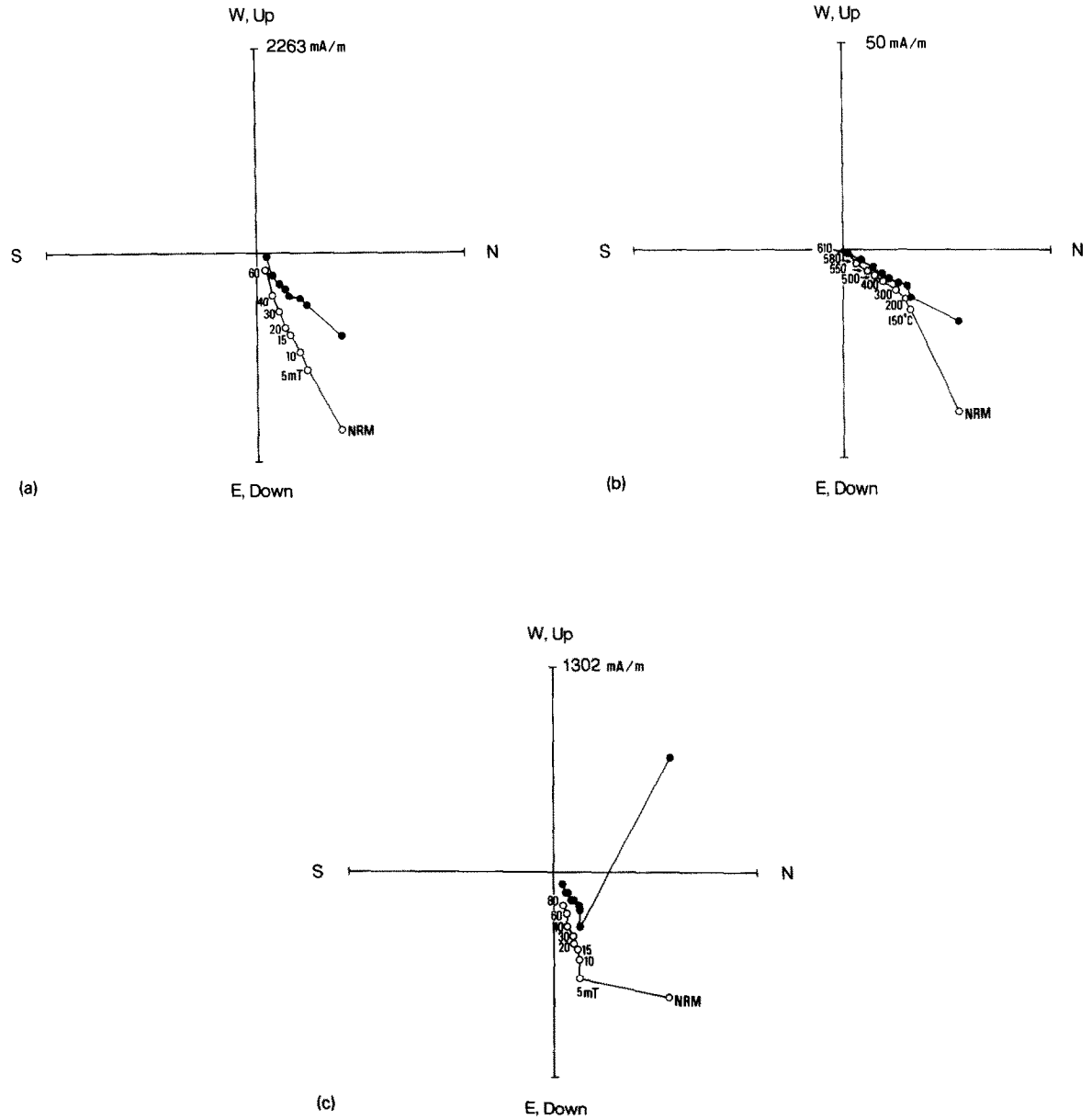


Fig. 3. Plot of stepwise demagnetization data. (a) Typical example of AF demagnetization for stable remanence data (sample KK 48-332). ● and ○ represent horizontal and vertical components, respectively. (b) Typical example of thermal demagnetization for stable remanence data (sample KK 20-B2). (c) Typical example of AF demagnetization data including an unstable secondary component (sample KK 26-178).

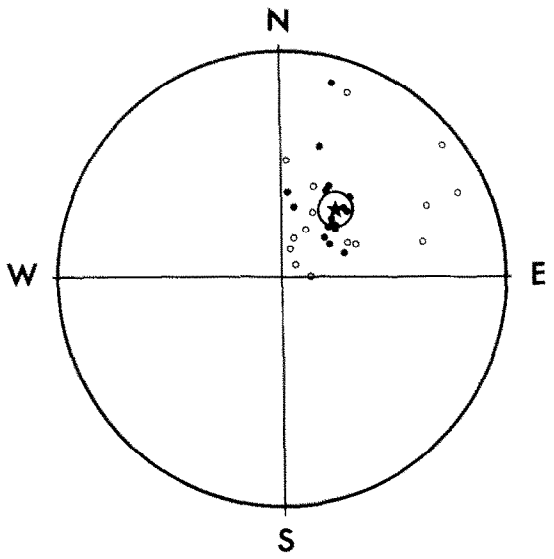


Fig. 4. Site mean paleomagnetic directions. Also shown is the overall mean direction (★) with the 95% confidence circle. Negative polarity samples (○) are reflected through the origin to an antipodal position.

herein are from the same volcanic field and are of similar age as those in Kim et al. (1986), their results are also listed in Table 2.

4. Results and discussions

The mean direction for the 25 flows is $D = 39.4°$, $I = 46.2°$ with $\alpha_{95} = 9.5°$, which differs statistically from the present dipole filed direction at the 95% confidence level. Because these data are from volcanic rocks, it is important to determine whether or not geomagnetic secular variation has been adequately averaged. First, antipodal normal and reversed polarity sites exist within both data sets which are statistically indistinguishable at the 95% confidence level (normal sites $n = 14$, $D = 36.7°$, $I = 48.0°$, $\alpha_{95} = 9.2°$; reversed sites $n = 11$, $D = 223.0°$, $I = -43.4°$, $\alpha_{95} = 20.1°$). Second, the paleomagnetic directions of sedimentary rocks and tuffs from the same region (Kim et al., 1986) are statistically indistinguishable at the 95% confidence level from those of the volcanic flows. Third, an angular variance of virtual geocentric pole of the volcanic flows is 24.3° (+8.2°, −5.0° with 95% confidence interval, Cox, 1969) which is about the same range of the worldwide paleosecular variation dispersions at similar latitudes during 45–5 Ma (Merrill and McElhinny, 1983). Thus, we feel that the middle Tertiary mean paleomagnetic directions reported in Table 1 represent an accurate sampling of the geomagnetic field and are not solely the results of secular variation of the earth's magnetic field.

When these results are combined with those of Kim et al. (1986), we obtain a mean direction of $D = 40.1°$, $I = 47.4°$, $\alpha_{95} = 7.9°$) for 33 sites of Late Oligocene to Middle Miocene age. Although the mean inclination is slightly shallower (a few degrees) than the present geomagnetic dipole field inclination ($I = 55°$ for 35.8°N), the continuity of geology of the region argues against a

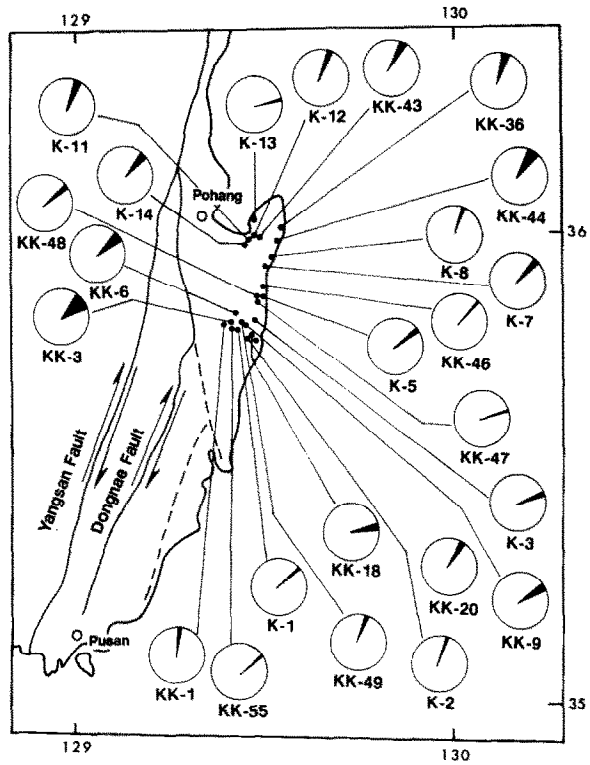


Fig. 5. Map showing Early–Middle Miocene paleomagnetic results from southeastern Korea. The circles show the direction of magnetic declination values (shown as mid-point of the darkened section) for each site. The size of the wedge corresponds to the α_{95} value. Negative polarity samples are reflected through the origin to their antipodal position.

significant northward drift in this region with respect to East Asia during the Neogene. The mean paleomagnetic declination deflected away from the dipole direction suggests that portions of southeastern Korea underwent a large clockwise rotation during the Neogene. Paleomagnetic vector rotations with vertical axes are commonly observed in regions of strike-slip faults where the rotations of the magnetic vectors are probably caused by the rotation of small crustal blocks caught between larger parallel fault sets in the region (Luyendyk et al., 1980; Ron et al., 1984; Nur et al., 1986). In the Pohang Basin, two large right-lateral faults (the Yangsan Fault and the Dongnae Fault, Fig. 1) are observed and both faults show evidence of recent activity. In addition, minor faults with trends similar to major faults occur in the field region (Fig. 1). Large earthquakes have occurred on these faults. These faults are mapped as cutting Early to Middle Miocene volcanics in the field region. Large right-lateral strike-slip faults and secondary faults in the study area suggest that the rotations discussed here are reflecting small crustal units caught up in the dextral shear of the region. We lack data from Late Miocene and Holocene rocks from the Pohang region and are unable to place a limit on the upper age of the rotation. As discussed above, paleomagnetic data from Pliocene to Holocene rocks from both Jeju Island and the Jeongog Basin (north of Seoul) are indistinguishable from the present dipole field direction. Because the rotations observed from the Pohang region are, we believe, of a local nature, their relationship with the data from either Jeju or the Jeongog Basin is not clear.

Table 2
Summary of paleomagnetic data from Korea and southwestern Japan

Age	Ref.	Location	N	D	I	α_{95}	VGP		A_{95}
							Lat. (°N)	Long. (°E)	
Miocene	a	(1)	25	39.4	46.2	9.5	56.7	217.2	9.1
Miocene	b	(1)	8	42.6	51.3	16.2	55.4	209.3	18.6
Miocene	a,b	(1)	33	40.1	47.4	7.9	56.5	215.2	7.9
Miocene	c	(2)	24	50.3	48.6	7.4	48.5	214.5	7.9
Cretaceous	d	(3)	27	28.1	57.7	3.5	67.5	202.8	4.5

N = number of sites; D = mean declination; I = mean inclination; α_{95} = radius of 95% confidence circle; A_{95} = radius of 95% confidence circle about VGP.
References: a = this study; b = Kim et al. (1986); c = Hayashida and Ito (1984), Otofuji and Matsuda (1983, 1984), Otofuji et al. (1983); d = Otofuji et al. (1983), Lee et al. (1987).
Locations: (1) Data from southeastern Korea; (2) Data from southwestern Japan; (3) Data from South Korea.

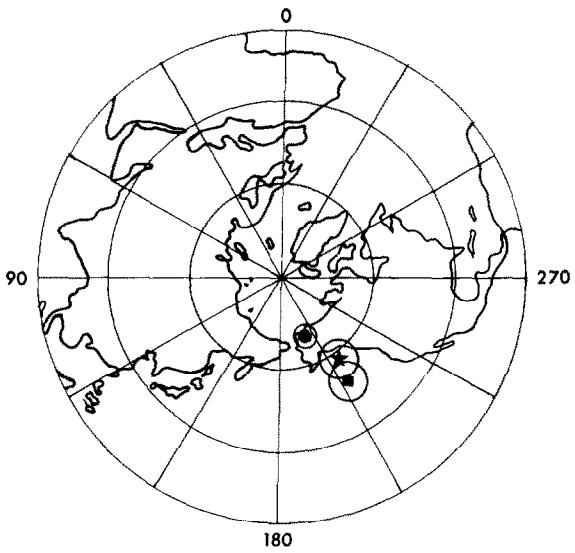


Fig. 6. Paleomagnetic poles from Korea and Japan. The Early–Middle Miocene pole from southeastern Korea is shown with ★. This direction is distinguishable from the Early Cretaceous pole (●) reported from South Korea (Otofuji et al., 1983; Lee et al., 1987) and indistinguishable from the Early–Middle Miocene pole (■) reported from southwestern Japan (Otofuji and Matsuda, 1983, 1984; Hayashida and Ito, 1984; Otofuji et al., 1985).

Our Early to Middle Miocene paleomagnetic pole for southeastern Korea (56.5°N, 215.2°E, α_{95} = 7.9°) is shown in Fig. 6. Also shown is the Early Cretaceous pole of South Korea (Lee et al., 1987) and the Early to Middle Miocene pole for southwestern Honshu (Otofuji and Matsuda, 1984). The Early to Middle Miocene pole for southeastern Korea is distinguishable from the Early Cretaceous pole for South Korea (Lee et al., 1987). We offer two possible explanations for this difference. The first explanation is that the Early Cretaceous pole represents the pole for the entire South Korean region, as suggested by Lee et al. (1987). In this explanation the Pohang re-

gion, east of the Yangsan Fault has rotated clockwise since the Middle Miocene with respect to the rest of Korea. This would locally displace the Early to Middle Miocene pole in a sense consistent with that shown in Fig. 6. The second explanation is that both the Early Cretaceous and Early to Middle Miocene data were collected from regions which have been rotated from the dipole field direction during the Neogene by movements along NE-trending right-lateral faults. We currently favor the first alternative because: (1) the data from South Korea are consistent with an Early Cretaceous pole for China, suggesting that it has shared a common translational history with China since Cretaceous times; and (2) the presence of the fold test in Early Cretaceous data (Lee et al., 1987) would argue that the area was rotated prior to folding.

The Early to Middle Miocene pole for southwestern Japan (Fig. 6) has been used to support the opening of the Japan Sea during the Miocene via a rigid block rotation (about 55°) of the southern Japanese Islands eastward from continental Asia. Although the Neogene rotations for southwestern Korea (Fig. 5) are not quite as large as those reported for southwestern Honshu, they are similar in direction. Recent data reported by Ishikawa and Torii (1988) show that Miocene samples collected from nearby Tsushima and Goto Islands (small islands between South Korea and Japan) are rotated counterclockwise. This suggests that the Miocene deformation in this region is more complex. We would simply note the close proximity of these two regions (400 km) and suggest that the cause of the rotations may be related to a similar tectonic event, if the rotations in southeastern Korea occurred during the same time period as those recorded in southwestern Honshu.

5. Conclusion

The data obtained allow us to conclude that since the Early Miocene southeastern Korea has undergone clockwise rotation of about 40°. The observed rotation is probably related to a large lateral shear couple in this region during the Neogene. This shear was considered to be taken up by the Yangsan and other right-lateral faults in Korea. As a result of this dextral shear, small crustal blocks in southeastern Korea were rotated in a clockwise sense. The paleomagnetic rotations observed in southwestern Japan and southeastern Korea are similar in magnitude and direction. This suggests that the rotation in both areas may be related to a similar tectonic event during the Neogene.

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